

ST 3011 – Statistical Programming
Individual Assignment – Semester 2 AY2025/26

Total: 100 marks

Due by 11:59 PM on 10th March, 2026

Instructions:

- In your previous group project, you conducted a statistical analysis using Python. For this individual assignment, you will extend that work by transforming it into an interactive web application using **Streamlit** (recommended).
- Your task is to develop a user-friendly application that enables users to explore the dataset, run statistical models, and interpret results through an interactive interface.
- This project must be completed **individually**, even though the original analysis was done in a group.

Learning Outcomes:

By completing this assignment, you should be able to:

- Translate a statistical analysis workflow into an interactive application.
- Communicate statistical results clearly to non-technical users.
- Produce well-structured and well-documented Python code.
- Critically reflect on the limitations of your model and propose improvements.

Core Requirements:

Your web application must include the following components.

- **Data Component**
 - Use the dataset from your group project. The application should load and perform any required data cleaning or preprocessing.
 - Provide user interaction with the dataset (e.g., filters, selections, or input controls).
- **Modelling Component**
 - Your app must implement at least two of the following: Regression, Classification, Clustering, Hypothesis testing, Time series analysis, Simulation.
 - One analysis may be from the previous group project.
 - One analysis must be newly introduced and not part of the previous group project. This will be assessed as the individual contribution.
- **Interactive Visualisations**
 - You must include at least two interactive visualisations.
 - Provide user-controlled options such as filters, sliders, or parameter inputs.

- **Interpretation**

- Each analysis or visualisation must be accompanied by a clear interpretation in simple language.
- Clearly state assumptions and limitations where relevant.

Deliverables:

- **Web Application:** A working Python web app that runs locally.
- **Individual Contribution Report:** A single page report describing the additional analysis that was introduced. Include a screenshot of that individual contribution.
- **Video Demonstration:** Prepare a short video (<5 minutes) demonstrating your app.
 - Show the app in action, including the models, visualisations, and interactivity.
 - Explain your individual contribution and the interpretation of results.
 - Highlight key features and usability aspects.

Submission Details:

- Zip up all necessary files to run the app locally and the report into one folder and name it with your index number. Submit the file to the submission link on LMS.
- Save the video demonstration in your Google Drive and share the link through LMS.

Marking Scheme: (Total 100 marks)

Criteria	Marks
Functionality and Usability (interactivity & responsiveness)	20
Data Component	10
Modelling Component (both models work correctly)	10
Individual Contribution (originality, justification & relevance)	15
Interactive Visualisations	10
Interpretations	10
Code quality	15
Video demonstration	5
Bonus (Exceptional creativity and innovation)	5

Milestone Guidance

Week 1	Week 2	Week 3	Week 4
app skeleton + data loading	first model working + plots	second model (individual contribution)	polish + interpretation + report + video

Focus on building a fully functional, clear, and interpretable application rather than striving for unnecessary complexity. A simple, reliable app with well-explained results will score higher than a complex application that is difficult to use or prone to errors.